# Earth Science and Geophysics Literature Compared with Basic and Applied Physics

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## Introduction/Background

*Earth science and geophysics papers often study open, messy natural systems, so they usually combine many kinds of data and explanation. Basic physics often aims for more general laws in more controlled settings, while applied physics is usually more focused on solving a defined technical problem. (LLM Memory)*

Earth science and geophysics sit in a different research setting than much of basic and applied physics. They often deal with large, complex, place-specific systems such as the Earth, oceans, atmosphere, crust, and interior, where direct control of variables is limited and observations are incomplete. Because of that, the literature often has to pull together field data, models, lab results, remote sensing, and past work to build a case. By contrast, basic physics often centers on isolating simpler systems and testing general principles under controlled conditions, while applied physics often targets device performance, engineering use, or specific practical outcomes. This difference in subject matter helps explain why earth science and geophysics papers often show more synthesis, more attention to setting and uncertainty, and broader use of sources from different methods and subfields. (Model-Generated)

## Epistemic complexity, synthesis, and triangulation

*Earth science and geophysics papers often have to make claims by combining several imperfect kinds of evidence rather than by relying on one clean experiment. Compared with much of basic physics, this makes the literature more synthetic, more method-mixing, and more focused on checking whether different lines of evidence agree. (9 sources)*

A main reason for the stronger synthesis in earth science and geophysics is that the objects of study are complex natural systems with limited experimental control, incomplete observability, and histories that matter for what we see now. Authors therefore often cannot validate a model in the simple way common in controlled sciences; instead, they build confidence by comparing field observations, experiments, simulations, theory, and multiple data types, while also checking whether the same story fits the system across space and time. Davis et al. stress that geophysics is “messy,” that models of complex natural systems cannot be validated in the usual sense, and that the field has had to develop new ways of testing models and theories (Davis et al., 2005). Kleinhans similarly describes geoscience as a mix of historical and experimental work, with frequent intermingling of present-day observables, experiments, simulation, and data analysis, because Earth systems preserve memory and show feedbacks, path dependence, and tipping behavior that do not reduce neatly to a few linear variables (Kleinhans, 2021) (Kleinhans et al., 2010) (Gramelsberger et al., 2020). This helps explain why geophysics is described as fundamentally multidisciplinary, using observational, experimental, and theoretical approaches together (Bezzeghoud et al., 2023), and why computation is treated as a core partner to field observation, lab analysis, experiment, and theory rather than a separate add-on (Hwang et al., 2017). In practice, that means earth-science papers often triangulate: for example, fracture studies can require coupled mechanical and geochemical reasoning and progress through observation, experiment, and modeling together, especially when reconstructing change through time (Laubach et al., 2019). Steventon et al. also note that geological reasoning often has to integrate quantitative and qualitative results, with expert interpretation helping researchers “read” the Earth (Steventon et al., 2021). Peng et al. add that geophysics often has huge amounts of data but still not the kind of complete data needed for analytic solutions, so it depends more heavily on information methods and technical tools to extract patterns and support inference (Peng et al., 2015). Compared with basic physics, then, the literature in earth science and geophysics more often advances by convergence across partial views than by a single decisive test; compared with applied physics, it is usually less about optimizing one device or process and more about integrating many methods to infer how a natural system works.

## Evidence

### (Davis et al., 2005)

Celebrating the Physics in Geophysics

> *"Because it deals with such complex systems, geophysics is rather messy, with little or no experimental control. It shares this last attribute with astrophysics, and more specifically cosmology, since there is only one Universe and only one Earth. Unlike cosmology however, the first inkling of physics-based prediction will be immediately confronted with hard data. So geophysics is a world of paradox and struggle because models of complex natural systems cannot be validated in the usual sense. Indeed, model validation by comparison of some output with realworld data does not confirm the underlying scientific conceptualization. There are cases of models that made accurate, quantitative predictions, but later proved to be conceptually flawed [3]"*

> *".Far from being an applied physics, geophysics has mutated the role of scientists who, in addition to remaining abreast in their traditional discipline, are now required to develop novel and still mostly undiscovered ways of validating their models and theories."*

### (Kleinhans, 2021)

Down to Earth: History and philosophy of geoscience in practice for undergraduate education

> *"With the disciplinary breadth of the geosciences comes a great diversity of methodologies and epistemic practices. Historical science and experimental science are often contrasted (Darling-Hammond et al., 2019) and some disciplines are indeed historical in that the development of Earth over a part or its entire period of existence is reconstructed (Baker, 1996;Kleinhans et al., 2005). Yet, at least half of the earth-scientific disciplines focus entirely on inference of causal laws and of causes of specific phenomena, or methods and technology to accomplish this, and use that knowledge for explanations and predictions through application of physics, chemistry and biology and through data analyses of present and past observables, experimentation, simulation, and combinations thereof (Darling-Hammond et al., 2019)(Gramelsberger et al., 2020)(Kleinhans et al., 2010). The practice, as visible at EGU, shows frequent interactions and intermingling of experimental and historic approaches. One reason is that complex earth systems have some memory in the sense that the internal structure is preserved and its state is to some degree contingent on past conditions. Geoscientists gain understanding of system properties and dynamics, such as feedbacks, path-dependence, downward causation and tipping points, that are not meaningful from the perspective of classic linear causal relationships between a few variables"*

### (Kleinhans et al., 2010)

HESS Opinions On the use of laboratory experimentation

> *"Abstract. From an outsider's perspective, hydrology combines field work with modelling, but mostly ignores the potential for gaining understanding and conceiving new hypotheses from controlled laboratory experiments. Sivapalan (2009) pleaded for a question- and hypothesis-driven hydrology where data analysis and top-down modelling approaches lead to general explanations and understanding of general trends and patterns. We discuss why and how such understanding is gained very effectively from controlled experimentation in comparison to field work and modelling. We argue that many major issues in hydrology are open to experimental investigations. Though experiments may have scale problems, these are of similar gravity as the well-known problems of fieldwork and modelling and have not impeded spectacular progress through experimentation in other geosciences."*

### (Gramelsberger et al., 2020)

Philosophical Perspectives on Earth System Modeling: Truth, Adequacy, and Understanding

> *"We explore three questions about Earth system modeling that are of both scientific and philosophical interest: What kind of understanding can be gained via complex Earth system models? How can the limits of understanding be bypassed or managed?*

*How should the task of evaluating Earth system models be conceptualized?"*

(Bezzeghoud et al., 2023)

Editorial: Geoscientific visualization in solid earth geophysics

*"Solid Earth Geophysics research is fundamentally multidisciplinary in its constitution. A large variety of observational, experimental, and theoretical approaches is used to investigate the structure and dynamics of the Earth."*

(Hwang et al., 2017)

Software and the Scientist: Coding and Citation Practices in Geodynamics

*"In geodynamics as in other scientific areas, computation has become a core component of research, complementing field observation, laboratory analysis, experiment, and theory."*

(Laubach et al., 2019)

The Role of Chemistry in Fracture Pattern Development and Opportunities to Advance Interpretations of Geological Materials

*"Future advances will require new approaches. Chemical processes play a larger role in opening-mode fracture pattern development than has hitherto been appreciated. This review examines relationships between mechanical and geochemical processes that influence the fracture patterns recorded in natural settings"*

*"Progress is possible using observational, experimental, and modeling approaches that view fracture patterns and properties as the result of coupled mechanical and chemical processes. A critical area is reconstructing patterns through time. Such data sets are essential for developing and testing predictive models."*

(Steventon et al., 2021)

Reproducibility in Subsurface Geoscience

*"(Frodeman, 1995) reviews the methods employed in geological reasoning, where some aspects of geoscience are very much like the "basic sciences," objective, empirical, with certain and precise results (akin to physics), while other aspects are more interpretive and qualitative, where prior knowledge and experience provide a framework to "read" the Earth. The challenge of integrating these quantitative and qualitative results makes the Earth sciences both an intellectually stimulating discipline to study, but as we have seen also provides a challenge for implementing reproducibility."*

(Peng et al., 2015)

From Geophysical Data to Geophysical Informatics

*"However, unlike most of the other branches of physics, geophysics and planetary physics can hardly collect enough data to derive analytical solutions for their research although the amount of data collected by geophysicists is huge. The objectives of geophysical research are complex, and most of them, such as the crust, mantle, and Earth's core, cannot be observed effectively. Therefore, compared with other disciplines in physics, geophysical study needs more information methods and techniques to deal with its data and problems."*

## Contextualization and interpretive reasoning

*Earth science and geophysics papers usually need more context than basic or applied physics papers because the meaning of the data depends strongly on place, history, and system setting. This pushes the literature toward more interpretive reasoning, where authors explain how they read incomplete evidence rather than only reporting a clean measurement or model result. (4 sources)*

Another clear difference is how much earth science and geophysics depend on context to make a claim. In many cases, observations do not speak for themselves: their meaning depends on local setting, past conditions, and how different Earth processes interact. Kleinhans argues that geoscience often mixes historical reconstruction with causal explanation because Earth systems retain memory of earlier states and show feedbacks, path dependence, and tipping behavior that do not fit simple linear cause-and-effect stories (Kleinhans, 2021) (Gramelsberger et al., 2020). This means papers often spend more space than basic physics papers on describing site history, regional structure, boundary conditions, and alternative readings of the evidence. Steventon et al. make this explicit: some parts of geoscience look like the basic sciences, but other parts are more interpretive and qualitative, where prior knowledge and experience help researchers “read” the Earth, and the challenge is to integrate these quantitative and qualitative results (Steventon et al., 2021). Bordonskiy makes a similar point more directly, saying that geology and geography study very complex objects and therefore use not only formal knowledge but also intuition, qualitative characterization, personal experience, and broad cross-disciplinary thinking to find links among interrelated geospheres (Bordonskiy, 2022). Compared with basic physics, then, earth science and geophysics writing is usually less able to separate observation from interpretation and more likely to show the chain of reasoning that connects partial evidence to a plausible Earth history or process explanation. Compared with applied physics, the difference is also one of purpose: instead of mainly showing that a device works or that a method improves performance, earth-science papers more often have to justify why a given interpretation fits a specific natural setting and why competing interpretations are less convincing.

## Evidence

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### (Bordonskiy, 2022)

Ice 0: the experimental proof of its existence – the result of combining approaches used in radiophysics, geology and geography

*"Geography and geology investigate rather complex objects; therefore, they have elaborated their own methods and approaches based not only on knowledge but also on intuition and 'qualitative' characterization of the objects of study. Here experience and personal perception of the object by a specialist play an important role. At the same time, research in the area of physics (geophysics) allows substantiation of the conclusions made by geologists and geophysicists on the basis of fundamental knowledge. Geologists and geophysicists are inclined to 'object-oriented thinking' (characteristic of the early stage of development of classical physics) and often rely on tentative intuitive methods. In fact, geology and geophysics are parts of synergetic, i.e., the science investigating self-organization in nature. Synergetic operates rather complex non-linear processes and requires deep knowledge of not only geology but also physics, mathematics, chemistry, and biology"*

*".I have noticed an important aspect of my colleagues' work in the area of geology and geography: this is the absence of 'fear' of working at the interface of sciences, indicating that these specialists are characterized by the broad range of approaches to carrying out research. All the geospheres are inter-related; therefore, geologists and geographers have to find ties between them in studying different objects. For example, for a geologist biological processes are a natural geological force changing the geological environment. Therefore, a geologist conducting research should understand the main principles of the flow of biological processes in nature. Deep knowledge of physics, chemistry and biology by narrow specialists from different disciplines allows detection of*

*these ties in a more effective way, when the efforts are united. However, the barriers on the way to breakthrough achievements often emerge when specialized approaches are applied."*

# Citation patterns and broader influence

*Earth science and geophysics papers often sit in wider citation networks than many physics papers because they connect to climate, hazards, resources, and other public-facing issues. This can make their references more outward-looking and can give the field more reach into policy and other non-academic domains. (2 sources)*

Citation patterns also reflect the broader role of earth science and geophysics. Because the field often links basic process questions to issues such as climate, natural hazards, water, energy, and land use, its papers can be cited across a wider mix of literatures than work that stays inside a narrower physics problem space. One recent study reports that documents citing geophysics show about **24% higher policy influence**, which the authors link to the political importance of climate policy (Lee et al., 2026). That does not mean every geophysics paper has more citations than every basic or applied physics paper, but it does suggest a different pattern of influence: geophysics work is often pulled into policy, environmental assessment, and interdisciplinary debate in ways that many basic physics papers are not. At the same time, evaluating this literature often requires looking not just at the paper itself but also at its reference list and who cites it afterward, since both backward and forward citation links help show how authors synthesize prior work and how their claims travel into other communities (Zolotov et al., 2024). Compared with basic physics, then, earth science and geophysics literature is often more embedded in cross-field and public-facing citation networks; compared with applied physics, its influence is less tied only to technical uptake and more often tied to policy relevance, environmental decision-making, and societal context (Lee et al., 2026).

### Evidence

**(Lee et al., 2026)**

The selective use of physics knowledge in policy: how interdisciplinary physics bridges subfields and shapes policy influence

*"Conversely, documents citing geophysics are associated with approximately 24 percent higher policy influence, likely driven by the political salience of climate change policy."*

**(Zolotov et al., 2024)**

Empirical models of electron precipitations at the Earth’s ionosphere high latitudes: a review

*"The manual evaluation stage includes the analysis and generalization of papers' content as well as an investigation of references lists and the citing literature (citations and citing literature analysis)."*

# Writing style and source use

*Earth science and geophysics papers often use a wider mix of source types than many physics papers. Their writing also tends to spend more space linking those sources together, because the argument usually depends on combining prior results from different methods and settings. (1 source)*

Writing style and source use differ across STEM fields, and that includes differences in the kinds of scholarly sources authors typically draw on (Blackburn et al., 2016). In earth science and geophysics, this usually means papers cite and discuss a broader mix of materials than a typical basic-physics paper: field studies, maps, survey data, lab work, modeling papers, remote-sensing studies, and regional case literature can all matter in the same article (Model-Generated). That source mix shapes the prose. Compared with basic physics, where writing can often move quickly from theory or experiment to result, earth-science papers more often have to explain why these different source types fit together and how each one constrains the interpretation (Model-Generated). Compared with applied physics, where sources are often chosen to support method design, performance benchmarks, or device comparisons, earth-science and geophysics papers more often use sources to establish setting, prior events, competing interpretations, and limits on what can be claimed (Model-Generated). So the writing style is often more connective and integrative:

authors do not just report findings, but also show how they assembled the evidential base from diverse literatures and why that combination is needed for the specific Earth problem at hand (Blackburn et al., 2016) (Model-Generated).

## Evidence

**(Blackburn et al., 2016)**

Changing the Scholarly Sources Landscape with Geomorphology Undergraduate Students.

> *"While such courses provide a foundation for basic writing skills, the STEM disciplines differ quite substantially in writing styles, especially in the types of scholarly sources typically used."*